

# GPT Models And Their Evolutions

■ From Words to Wisdom: Exploring GPT and Its Evolution

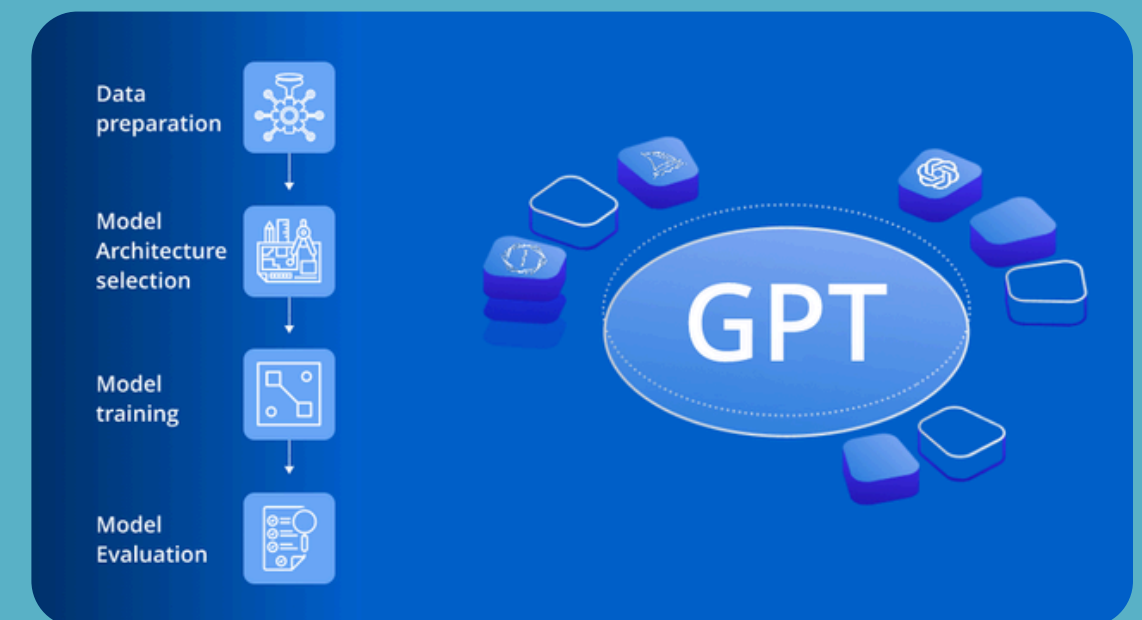
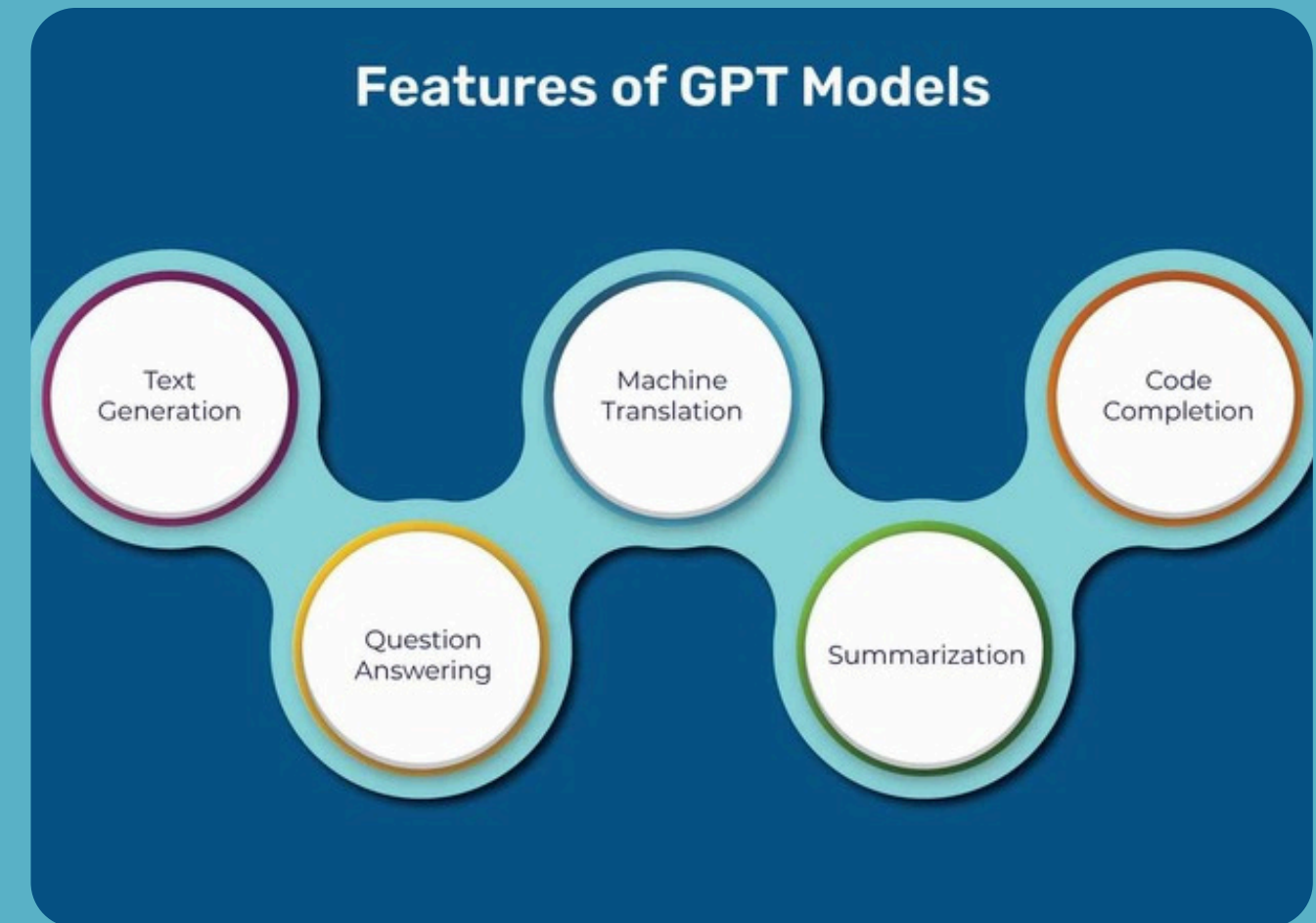
MLOM – IT3071



# Background & Intuition

- **NLP**
- **GPT like a student**
- **GPT stands for:**
  - **Generative**
  - **Pre-trained**
  - **Transformer**

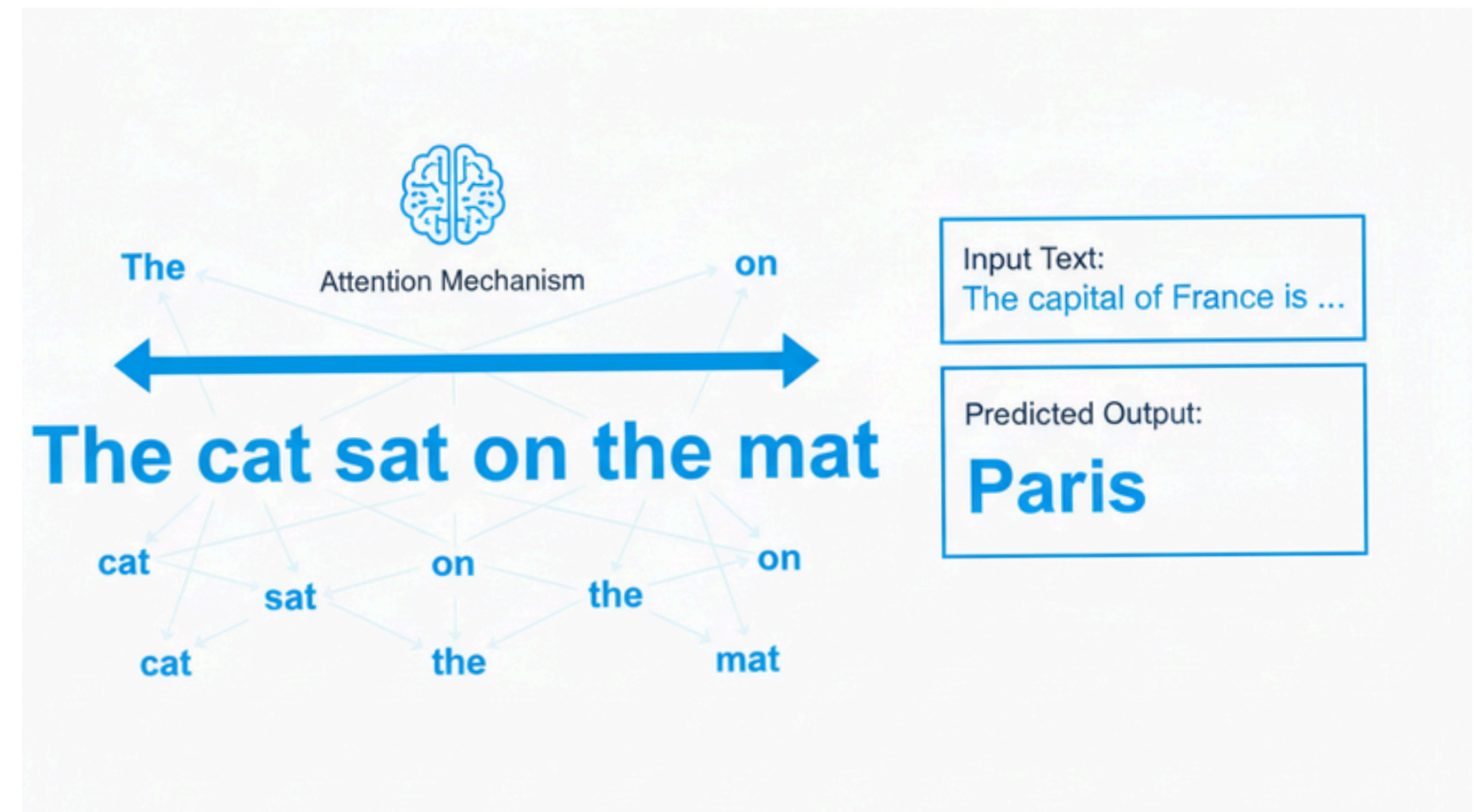
**GPT is a model trained to learn patterns in language and predict the next word.**





# How GPT Works

## Attention Mechanism



"GPT doesn't understand like humans — it predicts the next word using attention."

# Evaluation of GPT Models

## Early GPT Models (2018–2020)

- GPT-1 (2018): 117M parameters, proof of concept.
- GPT-2 (2019): 1.5B parameters, fluent text, initially withheld.
- GPT-3 (2020): 175B parameters, few-shot & zero-shot learning, essays/code.
  - One of the largest LLMs at the time

**The free version of ChatGPT is still powered by GPT-3.5, the most current version of GPT-3**





# ChatGPT & GPT-4 (2022–2023)



- **ChatGPT (2022)**

Based on GPT-3.5

Fine-tuned with RLHF → safer & interactive chatbot

- **GPT-4 (2023)**

Multimodal (text + images)

Improved reasoning, fewer hallucinations

**ChatGPT didn't just talk—it made the whole world listen**



# Future – GPT-5 & Beyond

- **GPT-5 (2025)**

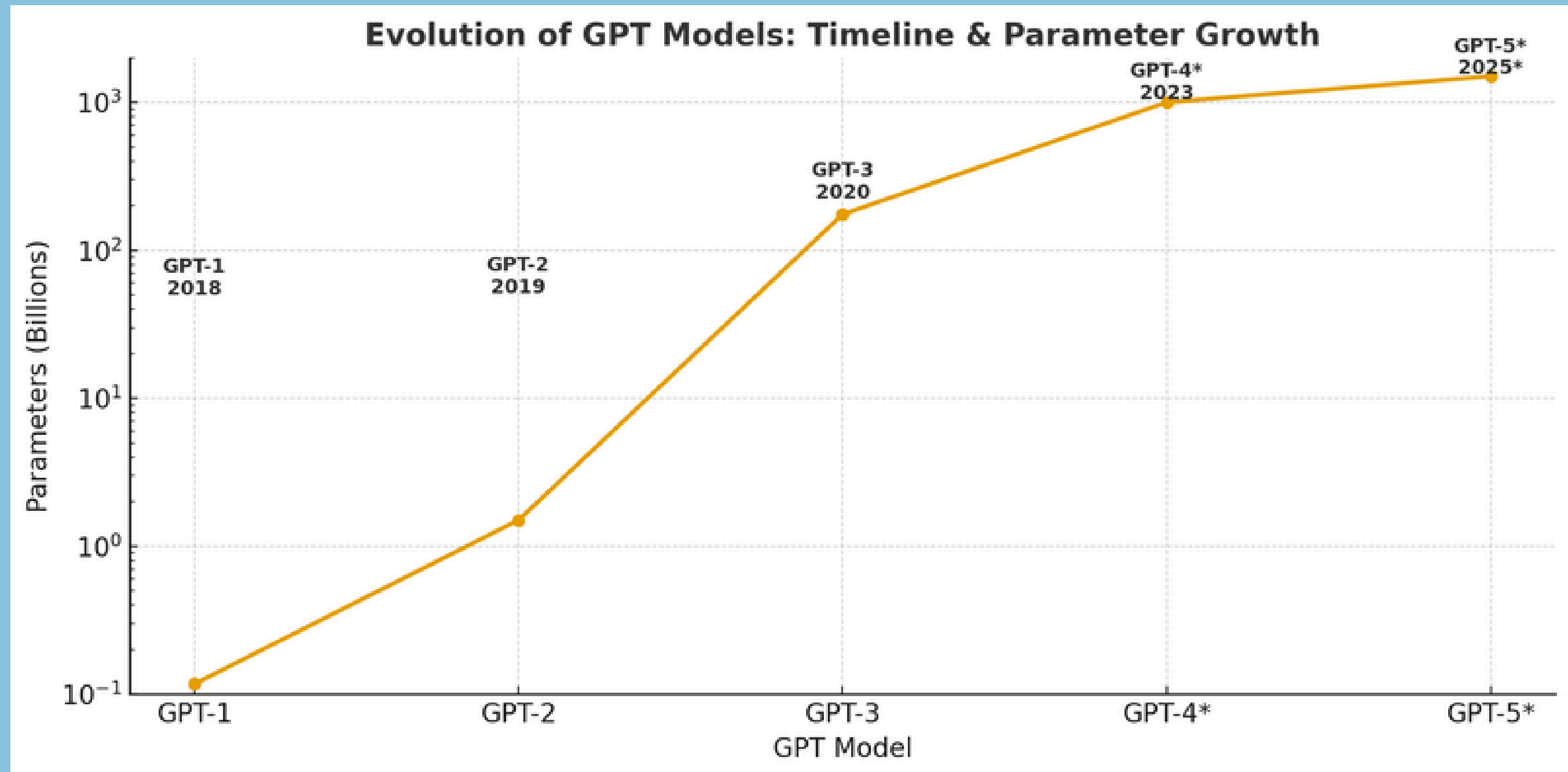
More efficient, smarter, safer  
Focused on reliability & reasoning

- Trend: From millions → billions → trillions of parameters
- ***Future = not just bigger, but more efficient & aligned***

***“The future of AI is not about being bigger,  
but being smarter and safer.”***



# Timeline & Growth of GPT Models



*“From millions to billions to trillions—the story of AI is exponential.”*



# Applications of GPT Models

💬 **Chatbots** – Answer questions & hold conversations

📖 **Education** – Act as personal tutors, guiding students step by step

🏥 **Healthcare** – Summarize patient notes quickly for doctors

💼 **Business** – Assist in writing emails & providing customer support

💻 **Coding** – Suggest code to help programmers



# Limitations & Ethical Concerns

## Limitations

✗ Hallucinations → gives wrong answers confidently.

⚖️ Bias → reflects internet biases in its training data.

💰 Cost → training requires millions of dollars and huge computing power.

## Ethical Concerns

📰 Misinformation → can spread fake news or enable plagiarism..

👷 Job Impact → fears of replacing human work.

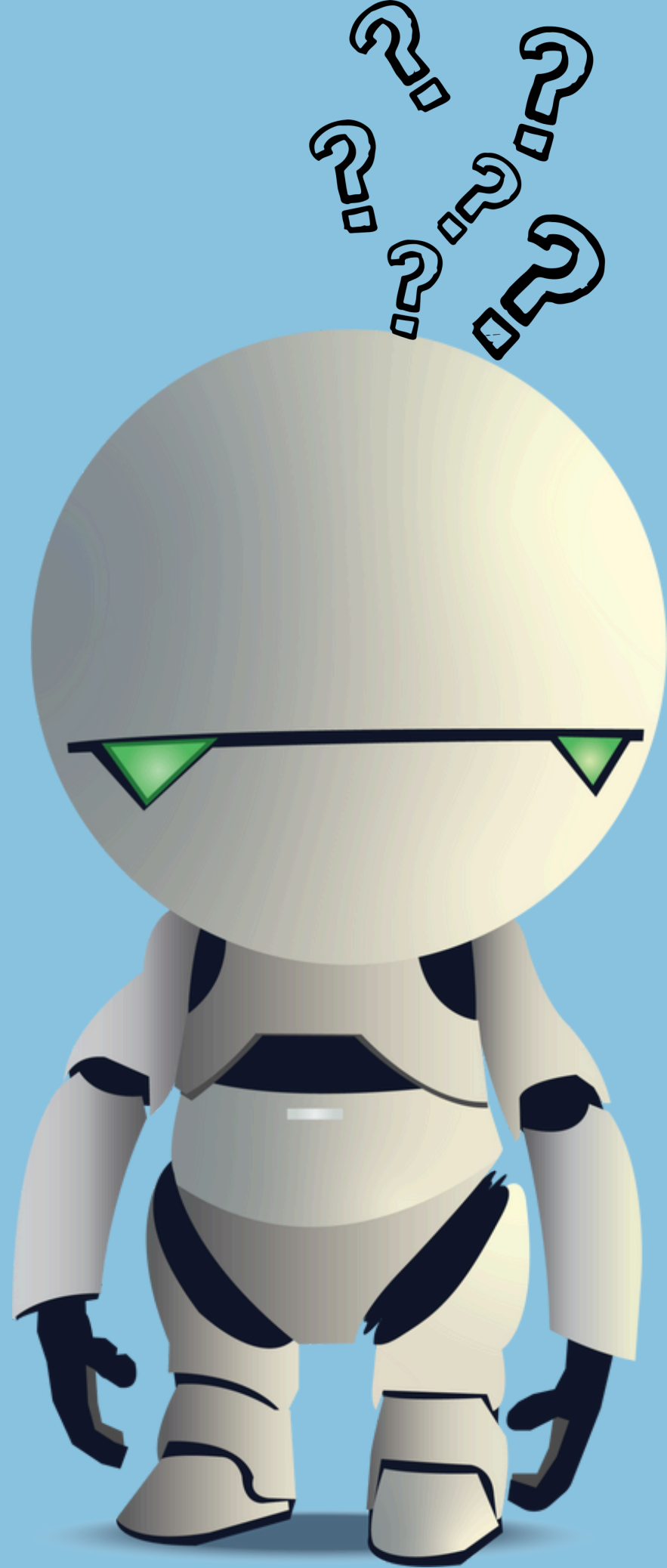
🌍 Environment → high energy use during training affects sustainability.

# Conclusion

🧠 GPT = Generative Pre-trained Transformer

📈 Evolution → bigger models, better training, smarter performance

🔮 Future → multimodal (text, images, audio, video) + more efficient & safer



Any Questions?